

Cloud LearnX

Mathematics

Circle Theorems — Practice Worksheet

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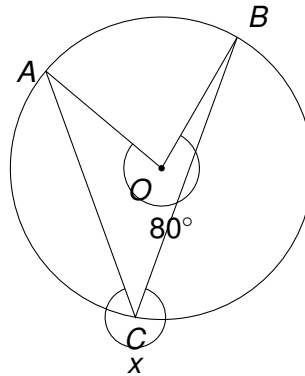
Topic: Circle Theorems

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.
Write your answers in the spaces provided.
You must write down all the stages in your working.

- 1 A , B and C are points on the circumference of a circle with centre O .



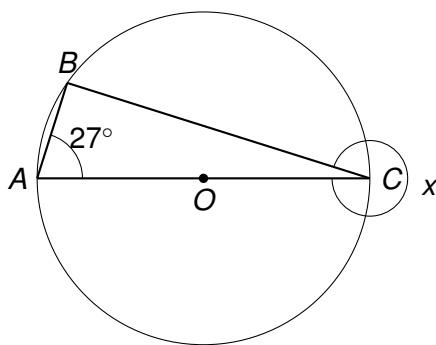
Angle $AOB = 80^\circ$.

Find the size of angle x , giving a reason for your answer.

$x = \dots\dots\dots^\circ$
(2)

(Total for Question 1 is 2 marks)

- 2 A circular badminton club logo is designed with points A , B and C on the circumference of a circle with centre O , where AC is a diameter of the circle.



Angle $BAC = 27^\circ$.

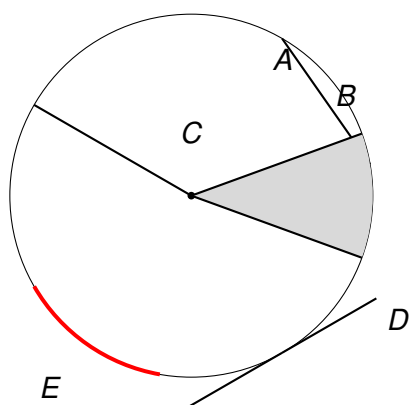
Find the size of angle x .

$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 2 is 2 marks)

3 A UI/UX designer is creating a circular badge icon for a Model United Nations app. The designer sketches the construction lines of the circle below, labelling five features *A*, *B*, *C*, *D* and *E*.



Using the diagram, write down the mathematical name for each of the labelled parts *A*, *B*, *C*, *D* and *E*.

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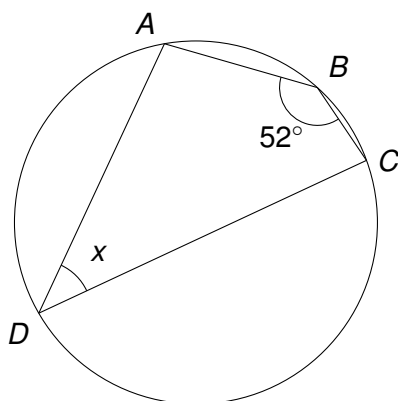
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(3)

(Total for Question 3 is 3 marks)

- 4 At a Model United Nations conference, delegates sit around a circular table with centre O . Four seats, A , B , C and D , are positioned on the edge of the table, in that order around the circle. Delegate lines of sight from B and D to the podium markers at A and C are shown below.



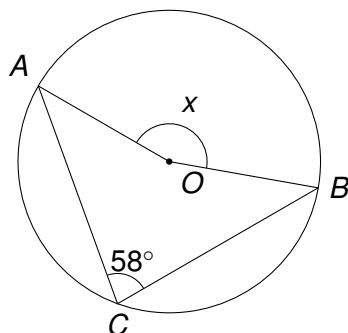
Angle $ABC = 52^\circ$.

Find the size of angle x , giving a reason for your answer.

$x = \dots\dots\dots^\circ$
(2)

(Total for Question 4 is 2 marks)

- 5 A stage designer for a Model United Nations conference is planning a circular seating arrangement with a central podium point O . Delegates sit at points A , B and C on the circular boundary, with O at the centre of the circle.



A , B and C lie on the circumference of the circle, and O is the centre.

Angle $ACB = 58^\circ$.

Work out the size of angle x , the angle AOB at the centre.

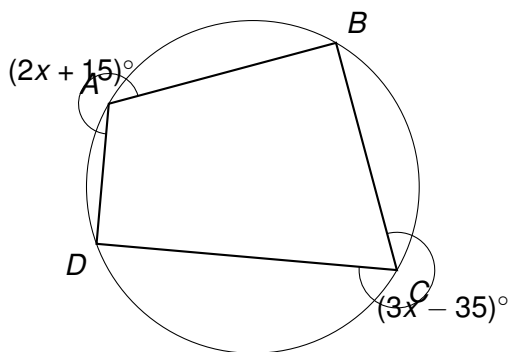
$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 5 is 2 marks)

- 6 At a Model United Nations conference, delegates sit around a circular table. Four delegates, A , B , C and D , are seated at points on the edge of the circular table, in that order, so that $ABCD$ is a cyclic quadrilateral.

Angle $DAB = (2x + 15)^\circ$ and angle $DCB = (3x - 35)^\circ$.



Work out the size of angle DAB .

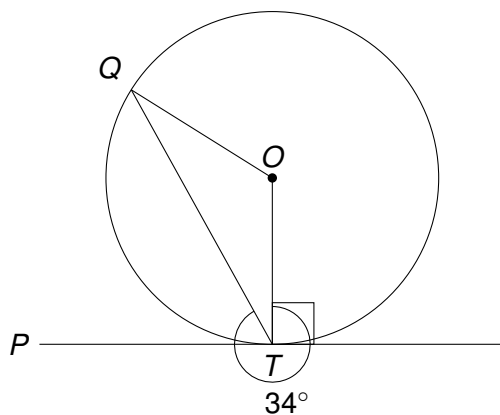
$DAB = \dots\dots\dots^\circ$

(3)

(Total for Question 6 is 3 marks)

- 7 A UI/UX designer is creating a circular badge icon for a badminton tournament app. The icon consists of a circle with centre O , a point T on the circumference where a straight edge (tangent) PT just touches the circle, and a chord TQ drawn from T to another point Q on the circle.

O , Q and T are positioned so that triangle OQT is isosceles, with $OQ = OT$ (both radii). Angle $OTQ = 34^\circ$.



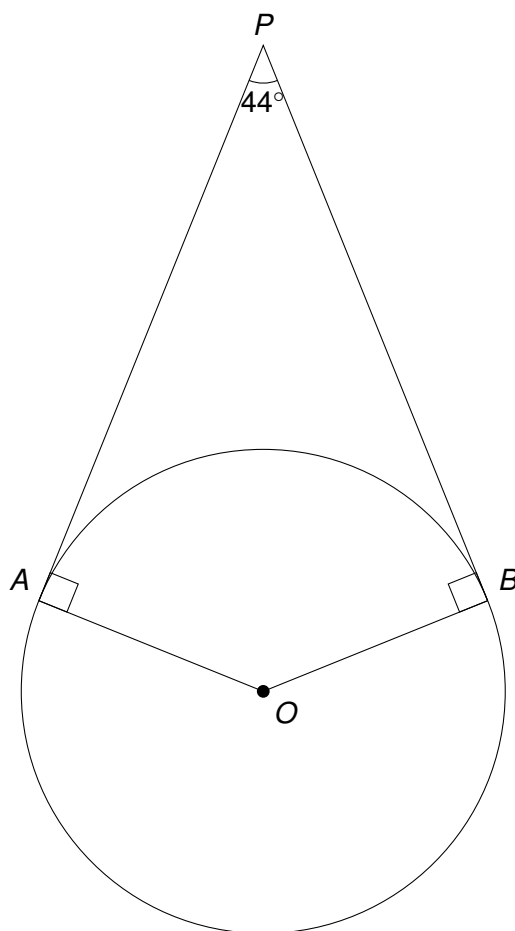
Find the size of angle QTP , the angle between the chord TQ and the tangent PT .

$QTP = \dots\dots\dots^\circ$

(2)

(Total for Question 7 is 2 marks)

- 8 A badminton club is designing a circular logo for its court banner. Two straight support cables, PA and PB , are attached to a circular hoop with centre O , touching the circle only at points A and B . The cables meet at an external point P .



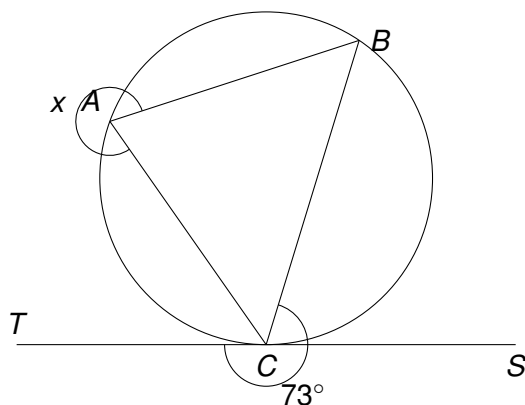
PA and PB are tangents to the circle. Angle $APB = 44^\circ$.

Work out the size of angle PAB .

$PAB = \dots\dots\dots^\circ$
(3)

(Total for Question 8 is 3 marks)

- 9 A circular badminton court marking is being designed using a compass and straightedge tool in a UI/UX prototyping app. Points A , B and C lie on the circumference of a circle, and ST is a tangent to the circle at point C .



Angle $TCB = 73^\circ$, where TCB is the angle between the tangent ST and the chord CB .

Find the size of angle BAC , marked x on the diagram.

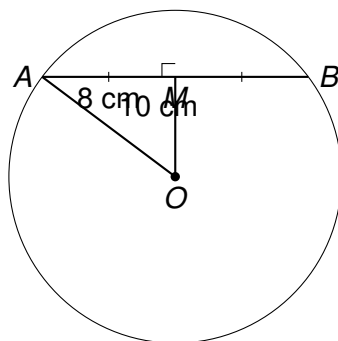
Give a reason for your answer.

$$x = \dots\dots\dots^\circ$$

(3)

(Total for Question 9 is 3 marks)

- 10** A UI/UX designer is creating a circular badge icon for an app. The badge is modelled as a circle with centre O and radius 10 cm. A straight decorative bar is drawn across the badge as a chord AB . The designer draws a line from O perpendicular to AB , meeting the chord at point M , where $AM = 8$ cm.



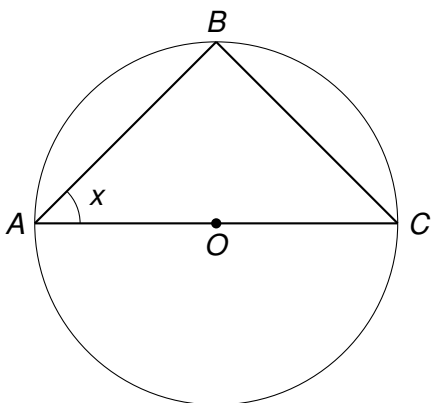
Calculate the length of OM , the perpendicular distance from the centre O to the chord AB . Give your answer correct to 3 significant figures.

$OM = \dots\dots\dots$ cm

(3)

(Total for Question 10 is 3 marks)

- 11** A UI/UX designer is creating a circular logo badge for a Model United Nations conference. The badge is modelled as a circle with centre O , where AC is a diameter. A decorative triangle ABC is drawn inside the circle, with point B on the circumference. To make the design symmetric, the designer ensures that $AB = BC$.



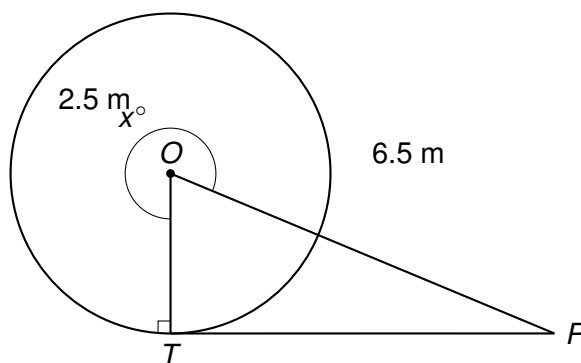
Given that $AB = BC$, show that $x = 45$.
Give reasons for each stage of your working.

(3)

(Total for Question 11 is 3 marks)

- 12** A satellite dish has a circular cross-section with centre O and radius OT . A support strut is bolted to the rim of the dish at point T , where the strut lies along the tangent to the circle at T . The strut runs from T to a fixing point P on the mounting pole, and a support cable connects O directly to P .

The support cable $OP = 6.5$ m and the radius $OT = 2.5$ m.



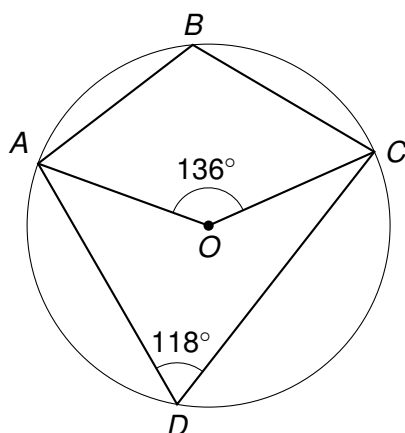
Work out the size of angle TOP , marked x on the diagram.
Give your answer correct to 1 decimal place.

$x = \dots\dots\dots^\circ$
(3)

(Total for Question 12 is 3 marks)

- 13** A logo designer is creating a circular emblem for a Model United Nations conference badge. Points A , B , C and D lie on the circumference of a circle with centre O , in that order around the circle. Lines OA and OC are drawn to the centre.

Angle $AOC = 136^\circ$ (the angle marked at the centre, on the same side as B), and angle $ADC = 118^\circ$.



- (a) Find the size of angle ABC .
Give a reason for your answer.

$$ABC = \dots\dots\dots^\circ$$

(2)

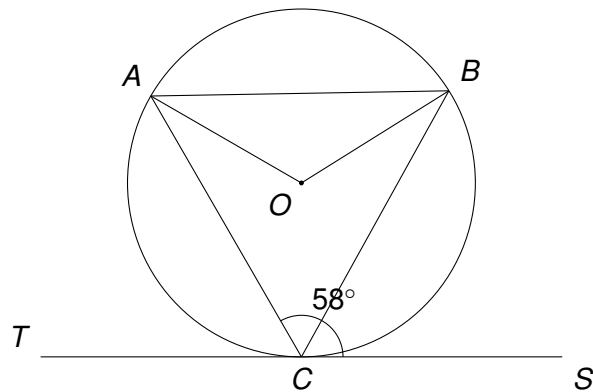
- (b) Work out the size of angle BCD .
Give a reason for each stage of your working.

$$BCD = \dots\dots\dots^\circ$$

(2)

(Total for Question 13 is 4 marks)

- 14** Points A , B and C lie on a circle with centre O . The line TCS is a tangent to the circle at C .



The angle between the tangent TCS and the chord CA is 58° , as shown.

- (a) Work out the size of angle ABC .
Give a reason for your answer.

$$ABC = \dots\dots\dots^\circ$$

(2)

- (b) Hence work out the size of angle AOB , where O is the centre of the circle.
Give a reason for your answer.

$$AOB = \dots\dots\dots^\circ$$

(2)

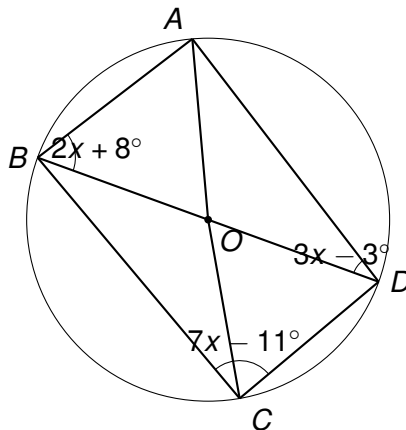
(Total for Question 14 is 4 marks)

15 A, B, C and D are four points on the circumference of a circle, with centre O , in that order around the circle. BD is a diameter of the circle.

O is joined to A and to C .

Angle $ABD = (2x + 8)^\circ$ and angle $ADB = (3x - 3)^\circ$.

Angle $BCD = (7x - 11)^\circ$.



Show that $x = 15$.

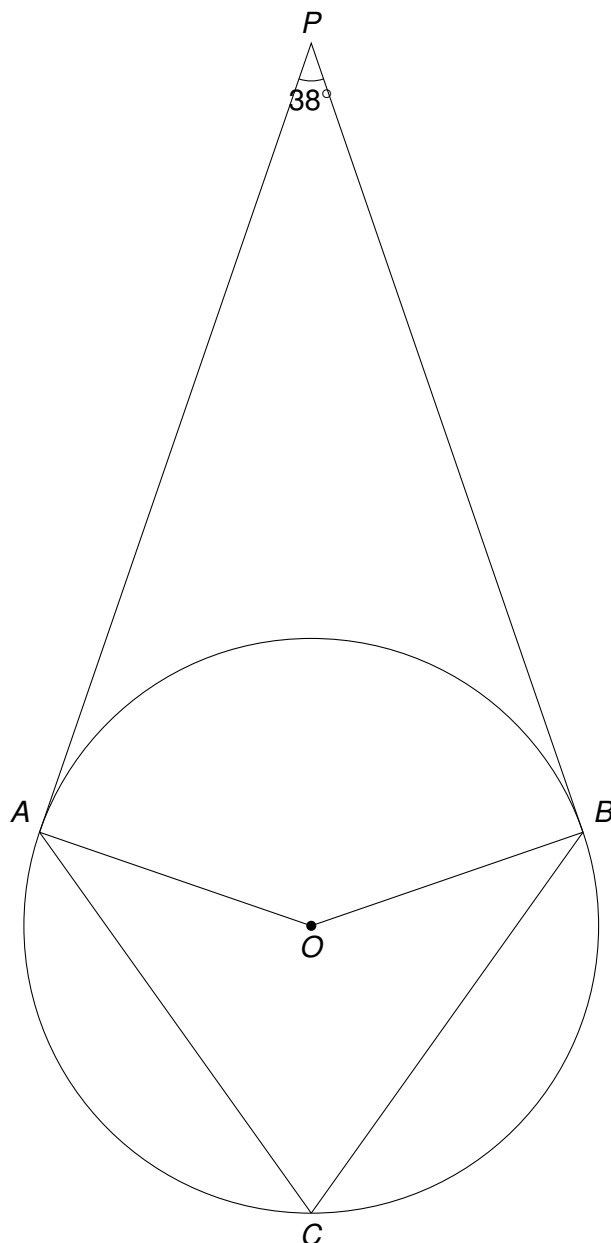
(4)

(Total for Question 15 is 4 marks)

- 16** A telescope's circular lens mount, with centre O , is held in place by two straight support brackets, PA and PB , which are tangential to the circular mount at points A and B . The brackets meet at an external mounting bracket point P .

A third fixing point C lies on the circumference of the mount, such that A , B and C form a triangle inscribed in the circle, with P , A , C and B arranged so that $ACBP$ is a quadrilateral (i.e. C lies on the major arc AB , on the opposite side of the circle from P).

Angle $APB = 38^\circ$.



- (a) Show that angle $AOB = 142^\circ$, giving reasons for each stage of your working.

(2)

- (b) Calculate the size of angle ACB , where C is a fixing point on the major arc AB (the arc on the opposite side of the circle from P).
Give a reason for your answer.

$$ACB = \dots\dots\dots^\circ$$

(2)

(Total for Question 16 is 4 marks)

TOTAL FOR PAPER IS ?? MARKS